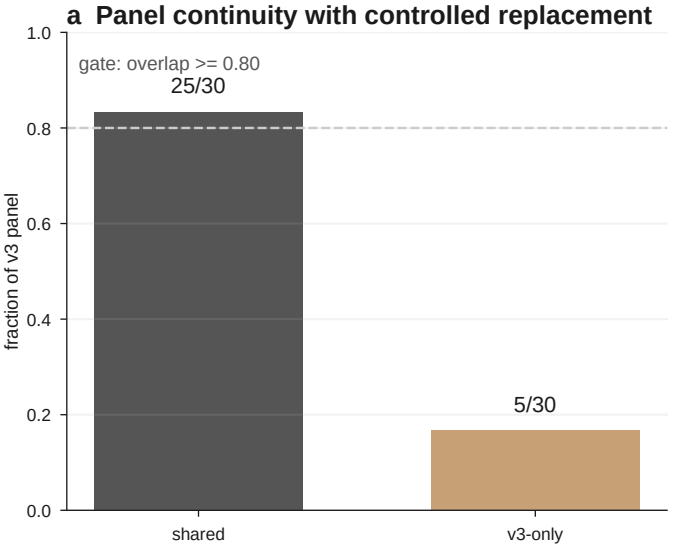
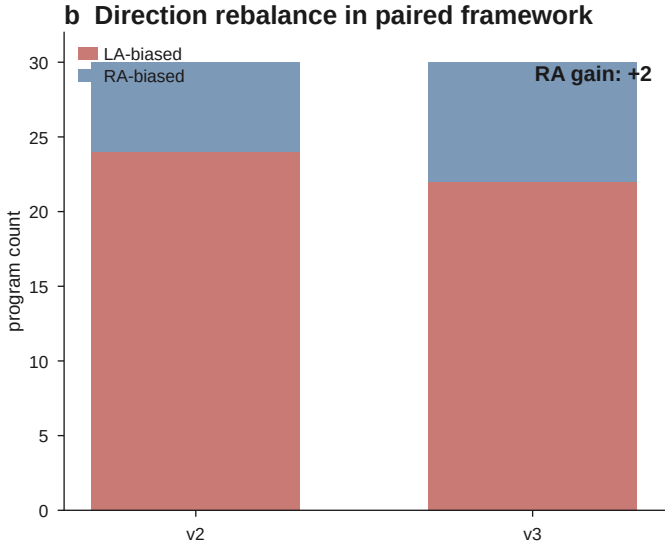
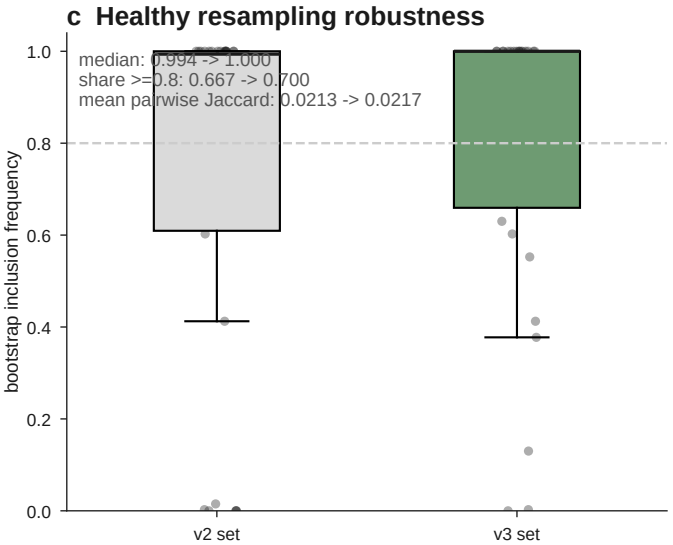
# Fig1D Healthy-only v2->v3 upgrade benchmark (paired framework)




## d Healthy-only upgrade gate audit

| | | |
|---|---|---|
| **Panel continuity**<br>rule: overlap_fraction_vs_v2 >= 0.80 | 0.8333333333333334 | PASS |
| **Direction balance**<br>rule: ra_count_v3 >= 8 and ra_gain >= 2 | RA 8 (gain +2) | PASS |
| **Redundancy control**<br>rule: mean_pairwise_jaccard_v3 <= 0.025 and delta <= 0.005 | 0.0217 (delta 0.0004) | PASS |
| **Bootstrap robustness**<br>rule: median inclusion freq >= 0.99 | 1.0 | PASS |
| **High-stability share**<br>rule: share(inclusion_freq >= 0.8) >= 0.70 | 0.7 | PASS |
| **Upgrade gain**<br>rule: v3 median and >=0.8 share >= v2 | median 0.994->1.000; share 0.667->0.700 | PASS |

**overall decision: upgrade-ready**

Metrics use only baseline panel tables, gene-set overlap, and healthy bootstrap resampling.